

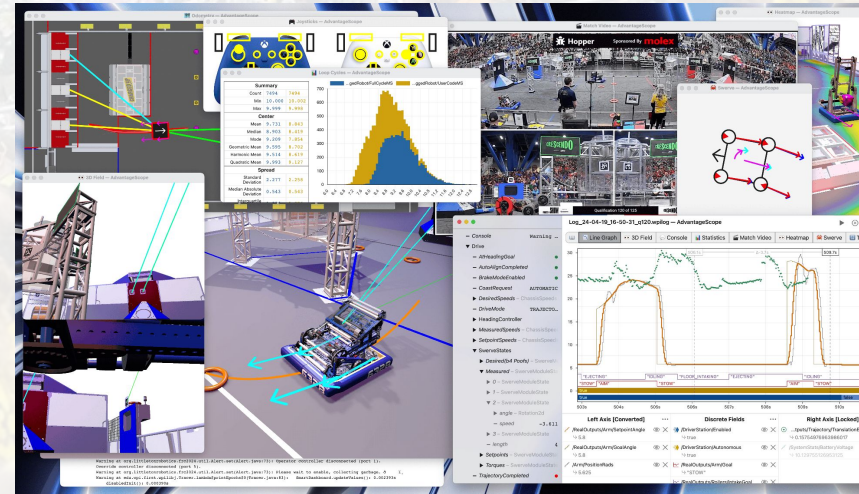


Software Tools

Written by Tim Ellis

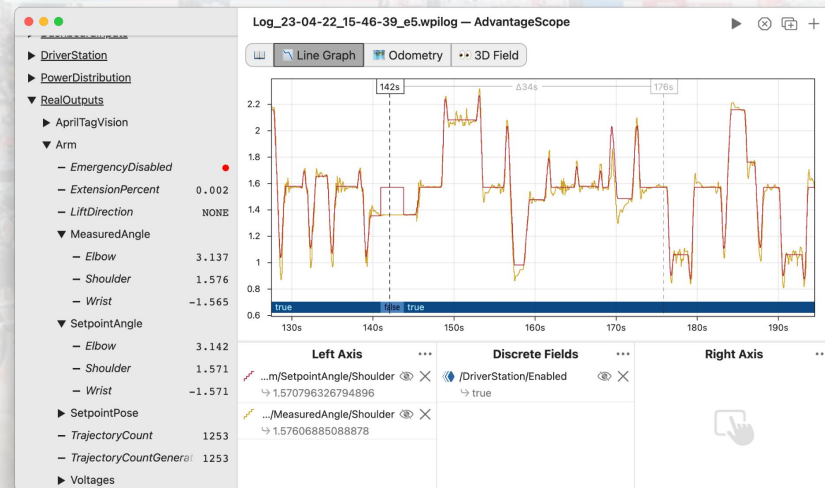
Advantage Scope

- Allows for advanced data visualization
- Data can be visualized in real time or after a match
- This can reveal important information about what the robot was thinking
 - Where did it think it was on the field?
 - Why did it shoot fuel at the ref?
 - What angle was the arm trying to go to when it broke?



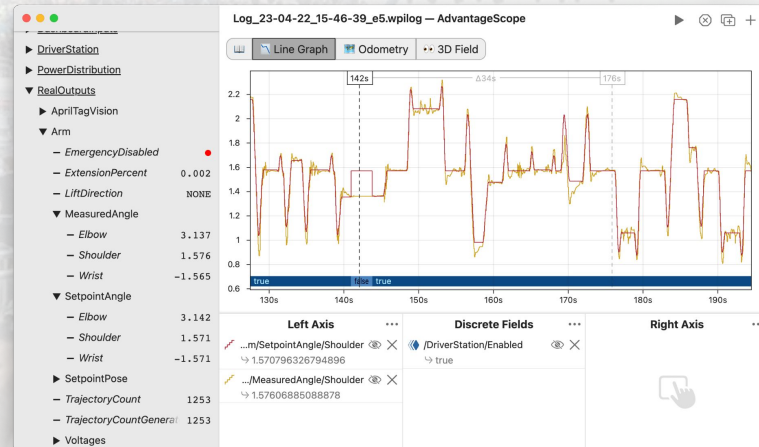
Advantage Scope

- The app is layed out similar to the file explorer on your computer
- On the left side of the screen, you will see a bunch of arrows, these are like folders
- If you see a bullet tick, then that is a 'field'
- A field is like a java variable, it has a name and a value



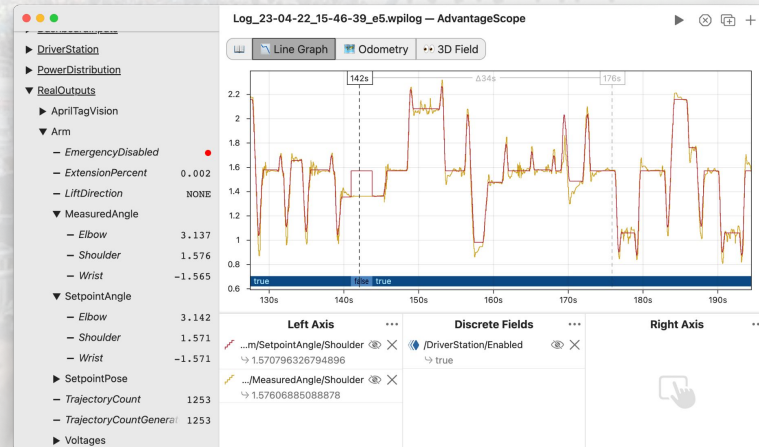
Advantage Scope

- There are several ways to visualize fields
- The most common is a line graph
- This allows you to see how one or multiple values changed over time
- To open a line graph, press the plus in the top right and choose “line graph”



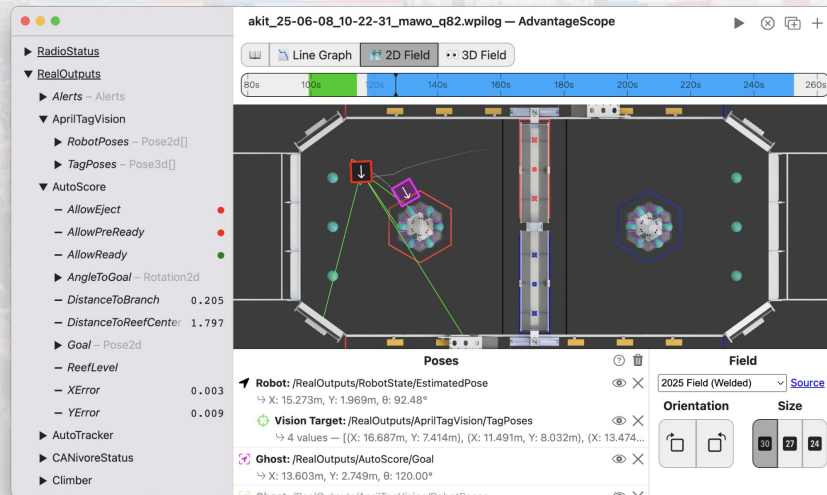
Advantage Scope

- You can also visualize discrete fields like boolean conditions
- To put a field on the graph, simply drag it from the left panel to the block you want it to be logged in (“Left Axis”, “Discrete Fields”, “Right Axis”)



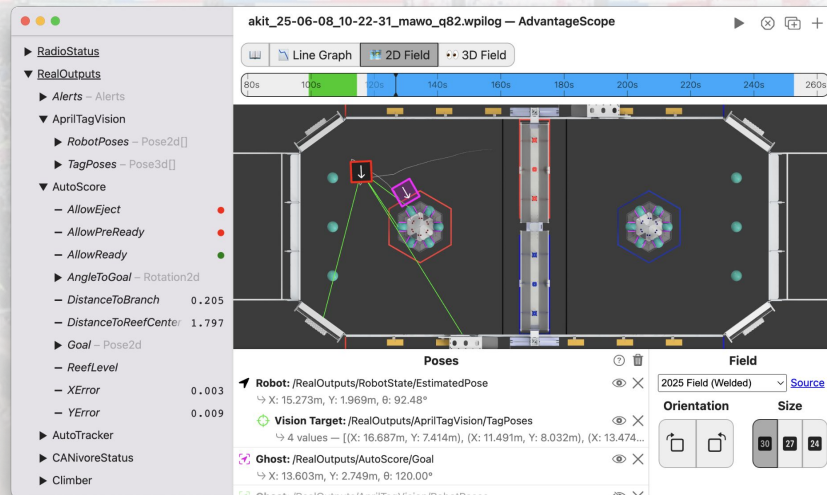
Advantage Scope

- Not all fields are numbers or booleans
- For example, the position of the robot is a Pose2d object
- To visualize a Pose2d, open a new tab by pressing the plus and choosing “2D Field”
- Here you can drag your pose2D and see it on the actual field



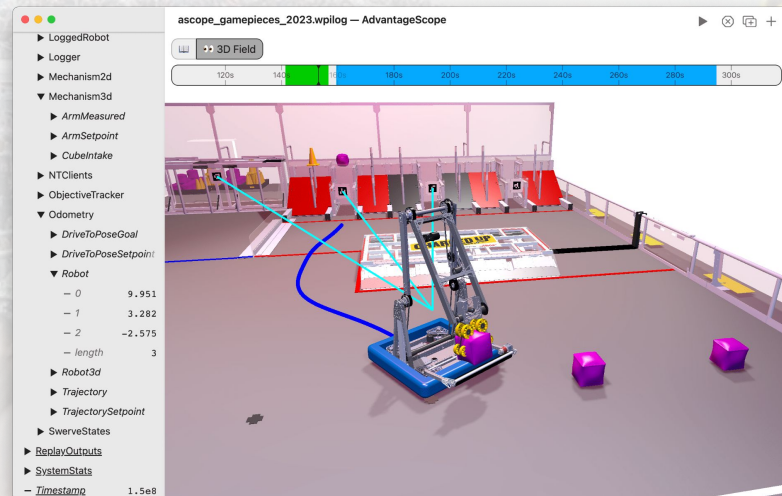
Advantage Scope

- To change what field you are using, there is a drop down on the bottom right
- You can also change the color and visibility of your robot by pressing the icon to the left of the word “Robot” in the Poses pane
- You can even visualize april tag targets to see what tags your robot is seeing!



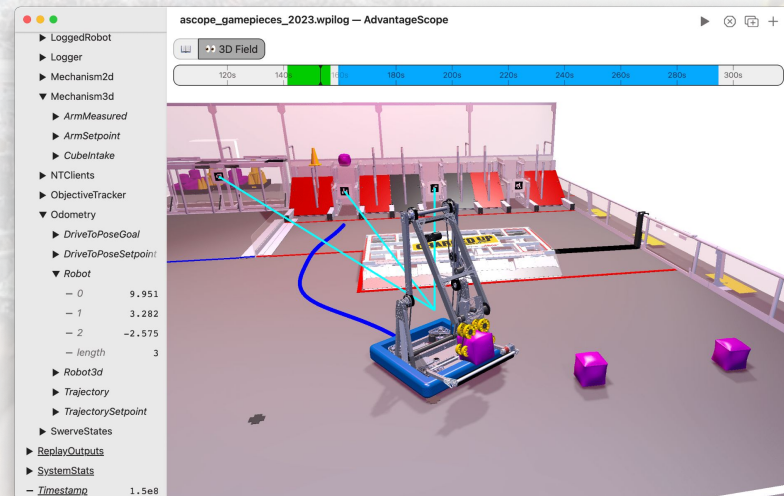
Advantage Scope

- Sometimes, a 2D Field can be less intuitive and more limited than a full 3D model of the field
- Open a 3D Field and drag your pose 2d in and you can see where your robot is in 3D
- You can even throw your CAD model into the app files and see your mechanisms moving with the robot



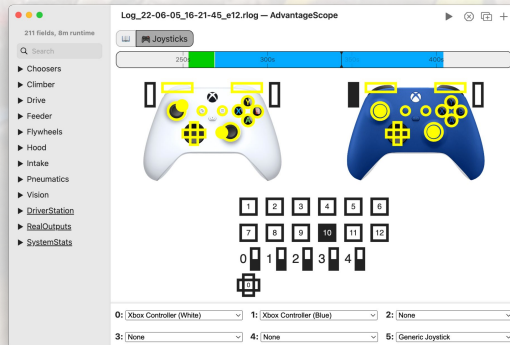
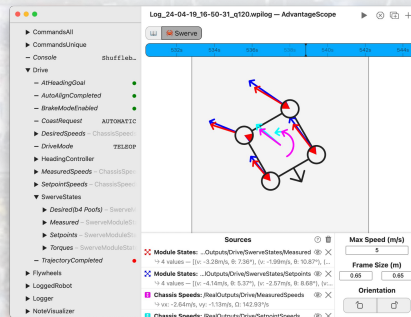
Advantage Scope

- Additionally, you can visualize game pieces and path trajectories
- Note that 3D field is not always a better choice than 2D field
- Sometimes seeing a quick top-down view can make things more clear



Advantage Scope

- There are several other ways to visualize information
- Please reference the Advantage Scope documentation to learn more!!!
- <https://docs.advantagescope.org/category/tab-reference>



	Line Graph	Alt+G
	2D Field	Alt+2
	3D Field	Alt+3
	Table	Alt+T
	Console	Alt+C
	Statistics	Alt+S
	Video	Alt+V
	Joysticks	Alt+J
	Swerve	Alt+D
	Mechanism	Alt+M
	Points	Alt+P
	Metadata	Alt+I

Advantage Scope



ADVANTAGE KIT

- Now that we know how to view information coming from the robot, how do we tell the robot to send out and 'log' that information?
- Great question students
- That is where Advantage Kit comes in
- Advantage Kit is not a standalone app
- Advantage Kit is a library that we use in our code to take numbers from Java to view them in real time and log them for later
- When we talk about getting values from code into AdvantageScope there's actually an intermediate step
- We send our Java values to NetworkTables

Advantage Scope

- As the name implies, NetworkTables is a table of values on the network
- This is the network between the robot and computer
- Think of it like a shared bulletin board where the robot can put notes (fields) up and apps on your computer can read those notes



Advantage Scope

- Currently, how we do this is with the `Logger.recordOutput` method
- This method takes two arguments
 - The name of the field (a String object)
 - The information to be logged (any data type)
- Every time the method is called, it will update that field

```
String PrimeShootString = "PrimeShootCommand";

Logger.recordOutput(PrimeShootString + "/hoodInPosition", elevationInPosition);
Logger.recordOutput(PrimeShootString + "/rotationInPosition", turretInPosition);
Logger.recordOutput(PrimeShootString + "/wheelAtPosition", wheelAtVelocity);
Logger.recordOutput(PrimeShootString + "/checkExtenderPosition", checkExtenderPosition);
Logger.recordOutput(
    PrimeShootString + "/actualTurretPosition", shooter.getTurretRotation().getDegrees());
```

Advantage Scope

- We can also log subsystem data with the @AutoLog annotation
- When we put @AutoLog before our Inputs class in our subsystems, it automatically logs all the information therein under the variable names

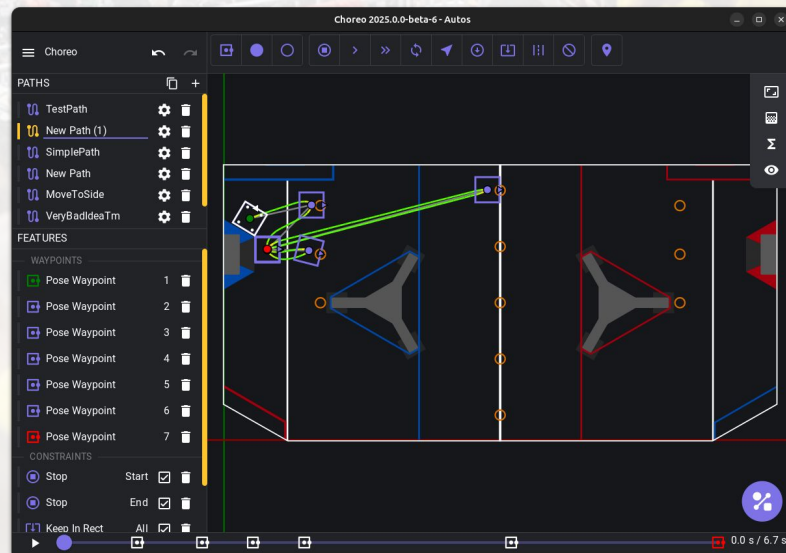
```
@AutoLog
class IntakeInputs {
    public double rollersVelocityRPM;
    public double rollersAppliedOutput;
    public double rollersCurrentAmps;
    public double rollersTempCelsius;
    public double extenderAppliedOutput;
    public double extenderCurrentAmps;
    public double extenderTempCelsius;
    public Rotation2d extenderPosition;
}
```





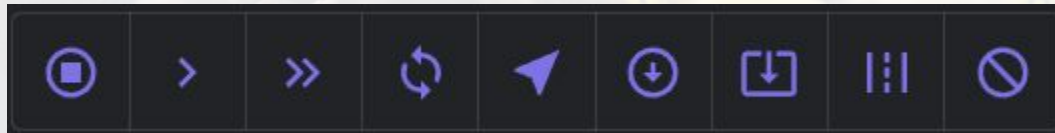
Choreo

- You may have heard of PathPlanner, Choreo is similar
- Choreo is an app where you can draw paths for your robot to follow on the field
- To make paths, you simply place waypoints on the field in the order you want your robot to reach them, generate the path, and Choreo will fill in all the intermediate points to make a smooth path



Choreo

- Sometimes you will have constraints on your path
- For example, you may want the robot to stay rotated at a certain angle, or keep out of a certain area of the field, or not exceed some speed, or not exceed some acceleration, or come to a full stop at a waypoint, or take a perfectly straight line between two points
- These can be achieved by adding Choreo constraints to different waypoints of your path



Choreo

- Making good paths in Choreo takes time and practice
- I deeply encourage you to download the app, make some paths and get creative

Choreo

- As far as the code side of Choreo, it is similar to Advantage Kit where we install a library that allows us to read paths
- The Choreo app will generate a set of .traj files in the deploy folder that will contain the data of the paths

```
"name":"Path1",
"version":1,
"snapshot":{
  "waypoints":[
    {"x":3.771560668945313, "y":7.2506170272},
    {"x":0.8527971506118774, "y":7.210460186},
    {"x":0.4947472214698791, "y":6.711747646},
    {"x":0.4180220663547516, "y":5.791047573},
    {"x":2.1059720516204834, "y":4.870347976},
  ]
  "constraints":[
    {"from":"first", "to":null, "data":{"type":
    {"from":"last", "to":null, "data":{"type":
    {"from":"first", "to":"last", "data":{"t
    "targetDt":0.05
```

Choreo

ChoreoLib

2025.0.3

2025.0.3



To Latest



- The Choreo library allows us to read that information into Commands that the robot can run
- To convert files into commands, we use the AutoFactory class

```
autoFactory = new AutoFactory(  
    driveSubsystem::getPose, // A function that returns the current robot pose  
    driveSubsystem::resetOdometry, // A function that resets the current robot pose to the provided Pose  
    driveSubsystem::followTrajectory, // The drive subsystem trajectory follower +  
    true, // If alliance flipping should be enabled +  
    driveSubsystem, // The drive subsystem  
);
```

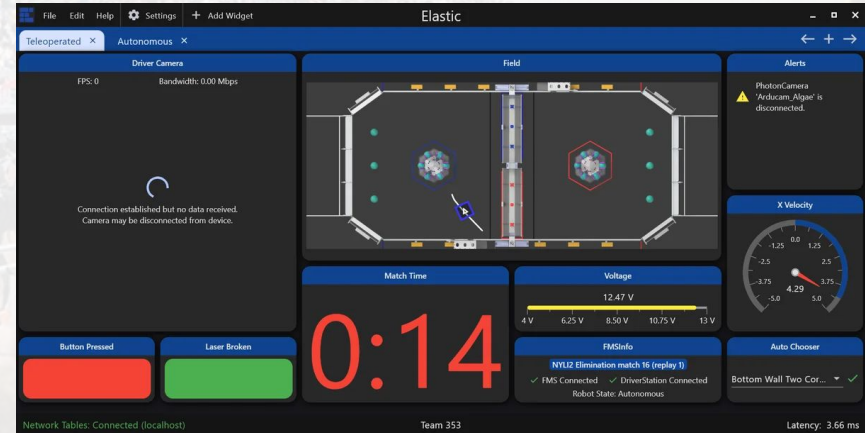
Choreo

- Once you create an AutoFactory, you can simply call the trajectoryCmd method with the path name as the parameter to create a command

```
// Follows deploy/choreo/myTrajectory.traj  
Command myTrajectory = autoFactory.trajectoryCmd("myTrajectory");
```

Elastic

- Elastic is the dashboard the drive team has up during a match
- Elastic can pull from the same NetworkTables as AdvantageScope
- However, AdvantageScope is geared towards data analysis while Elastic is geared toward real-time robot/match insights
- You can create buttons, timers, speedometers, fields, indicators, dropdowns, and more
- These are widgets that you can resize and move around the screen
- This is what we use to choose our auto



Phoenix Tuner X

- Phoenix Tuner X is the software we use to configure and interface directly with our motors
- Here you can config their names, CAN IDs, PID parameters, control mode, and more
- You can also run the motors with position control, velocity control, and voltage control
- Additionally, you can read their voltage, current, position, vel, accel, temp, and more
- This software works for all CTRE products including sensors like the Pigeon IMU and the CANRange sensor
- Fun Fact: you can download this as a phone app and control the robot with your phone (but please don't lol)



Conclusion

- You learned how you can visualize and analyze robot data in real-time or after the fact with AdvantageScope
- You learned how you can log data to NetworkTables with AdvantageKit
- You learned how to draw, create, and follow paths with Choreo
- You learned how to create dashboards for drive team with Elastic
- You learned how to interface directly with motors and sensors using Phoenix Tuner X
- But knowledge isn't understanding
- You won't have an intuitive sense of how to use these apps until you use these apps
- That's why it's important that you download these on your laptop and start trying to do things with them

Conclusion

- When you run into issues, your mentors will always be there to help
- However, there is very extensive documentation out there written specifically for students like you
- It would be very wise to take advantage of that:
 - Advantage Scope: <https://docs.advantagescope.org/>
 - Advantage Kit: <https://docs.advantagekit.org/>
 - Choreo: <https://choreo.autos/>
 - Elastic: <https://frc-elastic.gitbook.io/docs>
 - Tuner X: <https://v6.docs.ctr-electronics.com/en/stable/docs/tuner/index.html>
- Happy roboting!